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SURFBOARD PROPULSION SYSTEM

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EXECUTIVE SUMMARY

This project centers on the design and development of a surfboard propulsion system aimed at enhancing efficiency, safety, and user convenience. The initial concept featured a paddle, force springs, a rail system, a seat belt buckle, and a push-button activation mechanism. As the project progressed, the team identified functional and safety challenges, which led to significant design improvements focused on making the system practical and reliable for real-world use.

Early development efforts concentrated on optimizing the paddle's deployment angle to maximize propulsion force and water displacement, thereby increasing travel distance. However, based on advisor feedback, the team shifted focus toward improving energy storage and designing a robust locking mechanism capable of withstanding high loads. This change emphasized usability and ensured consistent performance in demanding environments.

To improve safety and operational efficiency, the team assessed various loading mechanisms. An initial block-and-tackle approach was rejected due to concerns about rope entanglement. Instead, a winch-based system was adopted, offering a more compact, cost-effective, and user-friendly alternative. Crucially, the winch included a built-in mechanism to prevent backward motion during reeling, allowing users to pause safely while resetting the paddle, enhancing overall reliability and ease of operation. The mechanical advantage provided by the winch was 26, which resulted in much easier loading operations than the previous design.

Further safety upgrades included the integration of a crossbow-inspired trigger mechanism to securely lock the paddle in place, along with a manual safety lock to prevent accidental deployment. The result was no accidental or unintentional paddle deployment, typical in the previous design.

The original HDPE paddle was replaced with a neoprene model housed in an aluminum frame. This design provided a higher drag coefficient of around 1.75 than the original flat plate's 1.28. The result was improved propulsion performance. The team upgraded to higher-strength springs with a spring constant of 375 N/m to enhance launch power, resulting in more powerful and efficient paddle deployment. The maximum speed of the board was observed to be 2.5 m/s, accelerated from a standstill, which corresponds well with the predicted models. However, one drawback of the new design is increased surfboard drag. The approximate new drag coefficient is 0.12 compared to the original 0.09.

1. PROBLEM STATEMENT

Surfing is a thrilling sport that connects individuals with the ocean, offering a unique sense of freedom and adventure. However, the physical demands of paddling into waves present significant challenges, particularly for beginners and individuals with limited upper body strength. Many first-time surfers struggle to catch waves, often leaving their initial sessions discouraged by the sport's difficulty. These barriers reduce the accessibility of surfing and prevent many potential enthusiasts from thoroughly enjoying its benefits.

To address these challenges, this project focuses on developing a surfboard propulsion system to provide additional thrust, enabling users to catch waves with less physical effort. The system targets beginners, individuals with physical limitations, and those seeking a more enjoyable surfing experience by reducing the strain associated with paddling. By integrating this propulsion system into surfboards, the project aims to make surfing more inclusive and accessible while enhancing the overall experience for users of all skill levels.

The mechanical propulsion system incorporates a paddle, a winch, springs, and a crossbow-inspired trigger mechanism. This setup delivers thrust by displacing water effectively, enabling the board to gain momentum without requiring excessive physical effort from the surfer. The design includes a locking mechanism to prevent backward motion during reeling and a manual safety lock to prevent accidental deployment. These features enhance the system's reliability and address potential risks, ensuring safe and consistent operation in various conditions.

In addition to its functional benefits, the design prioritizes sustainability and economic feasibility. Lightweight, corrosion-resistant materials are used to withstand ocean conditions while maintaining affordability. By addressing the physical, financial, and environmental challenges of traditional surfing, this project lays the foundation for a more inclusive and user-friendly approach to surfing, transforming the sport for a broader audience.

2. CONCEPTUAL DESIGN

The surfboard propulsion system results from iterative design and improvement based on lessons learned from previous prototypes. The project began with a thorough analysis of the previous design, identifying areas for improvement in efficiency, usability, and safety. One of the primary issues observed in the earlier model was the inefficiency of the paddle during deployment. In the previous design, the paddle only reached a 90-degree angle after traveling halfway along the rail, significantly reducing the thrust efficiency. To address this, our team initially explored methods to ensure the paddle reached a 90-degree angle as quickly as possible after release, thereby maximizing propulsion efficiency.

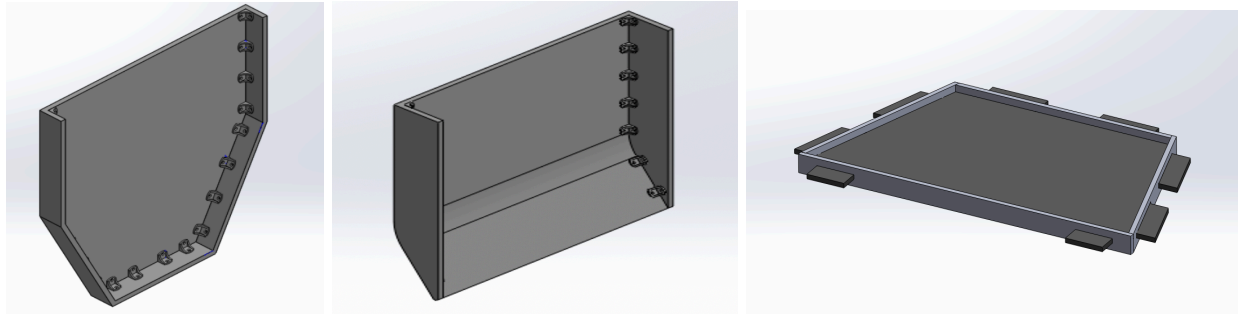


Figure 1: Paddle Design Iterations

Various paddle shapes and sizes were considered to improve deployment efficiency—initial ideas involved automatically opening paddles, which would use the force of drag to deploy. The team explored using magnets to facilitate smoother and faster paddle motion to aid this. However, after consulting with Dr. Basily, the magnet idea was abandoned due to concerns about their strength and durability in ocean conditions. Instead, a rocker arm mechanism was proposed to address this challenge. Rocker arms would be mounted along the rail and designed to lift the paddle into position when triggered, locking it securely in place. This concept aimed to improve paddle efficiency while maintaining structural integrity and reliability. While the rocker arms showed some promise in enhancing the system's efficiency, they failed to demonstrate a reasonable level of safety. In addition, it was assumed that the system must be loaded manually, not particularly catering to children and individuals with poor upper body strength.

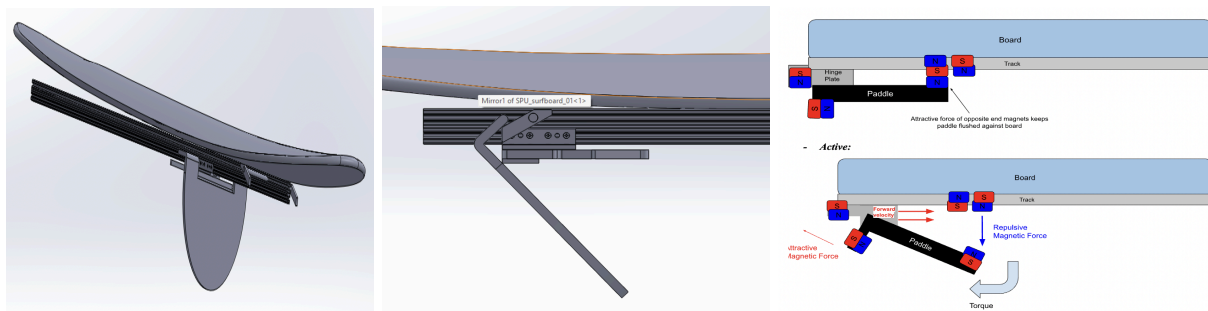


Figure 2: Rocker Arm Assembly and Magnet Concept

While the rocker arm mechanism showed promise regarding efficiency, the focus of the design shifted to safety, leading to the implementation of the trigger mechanism. Inspired by crossbows, this trigger system ensures the paddle remains securely locked until the user deliberately disengages the safety lock and activates the mechanism via a trigger lever. The seatbelt buckle, attached to the paddle, locks into the trigger mechanism to prevent accidental releases. The load is sufficiently large to cause harm to human beings, yet sufficiently small that the trigger can be made out of lighter material while maintaining safety. Dr. Basily recommended aluminum for the trigger mechanism due to aluminum's strength, weight, and corrosion-resistant properties. This design enhances safety and operational reliability, addressing a key concern from previous prototypes, which utilized a standard school bus seat belt system.

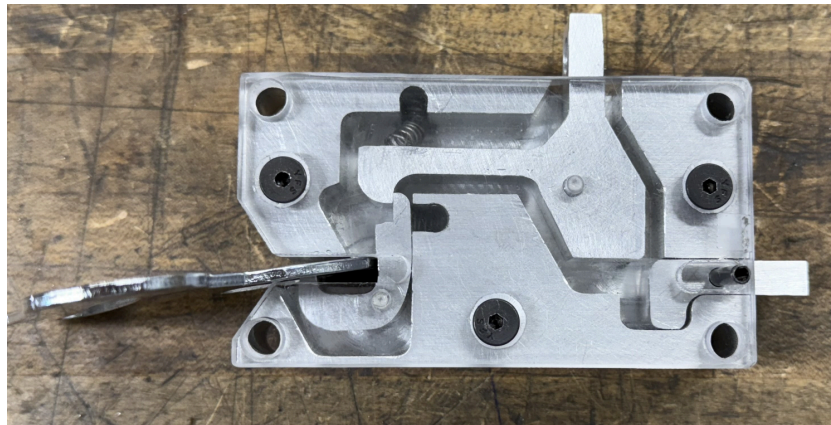
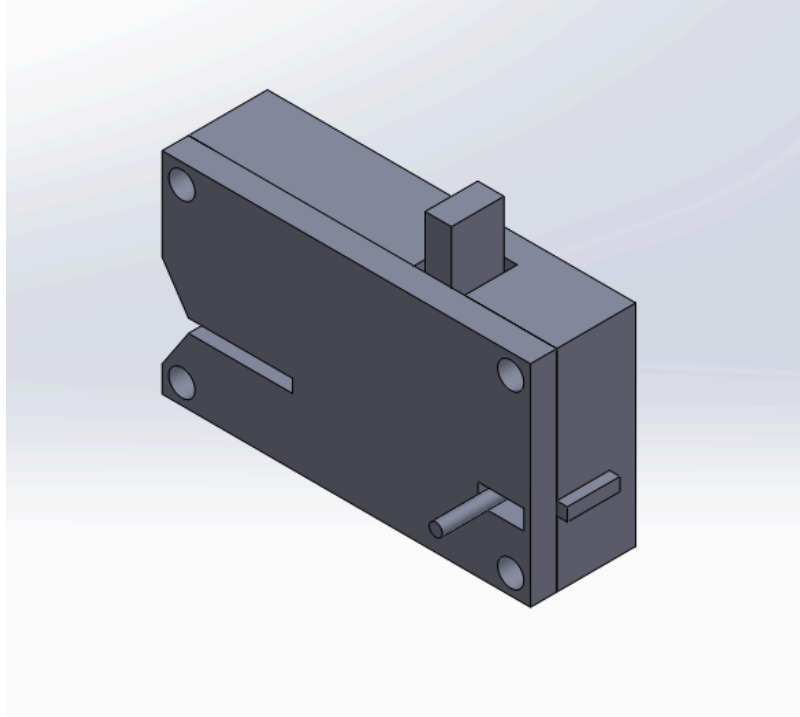


Figure 3: Trigger Mechanism

Another significant improvement involved the reloading process, which had previously relied on manual leg extensions to reset the paddle. While functional, this method was physically demanding and required the surfer to leave the board, limiting its practicality. As previously discussed, the physical demand required prevented the machine from achieving its goal of being easy for beginners to use. To overcome this, the team introduced a winch system that allows users to reload the paddle effortlessly while remaining on the board. This system features a high mechanical advantage to minimize the effort required to overcome high spring constants' resistance, enabling stronger springs for more energy storage and propulsion force. By securely mounting the winch to the surfboard, the system ensures stability during operation while eliminating risks associated with rope tangling—an issue with a block-and-tackle design that the group explored. Although the winch had numerous advantages, its main drawback was its

weight. The winch was produced out of stainless steel, which makes it quite heavy, shifting the board's weight toward the nose and increasing the possibility of drag during operation.

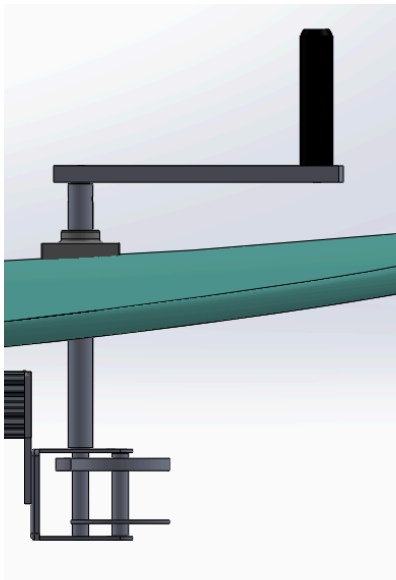
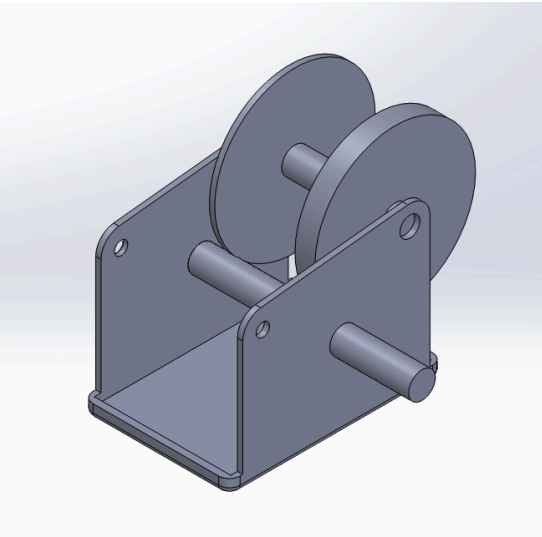


Figure 4: Winch Assembly

The lever release further simplifies the user experience, allowing surfers to activate the propulsion system conveniently. Positioned within the surfer's reach, the lever is directly connected to the trigger mechanism and releases the paddle upon activation. Once released, the paddle travels smoothly along the rail system, constructed from lightweight, corrosion-resistant aluminum to minimize weight and drag while improving longevity. The nature of the design ensures seamless integration with standard surfboards, maintaining their structural integrity and performance even when the propulsion system is not in use.

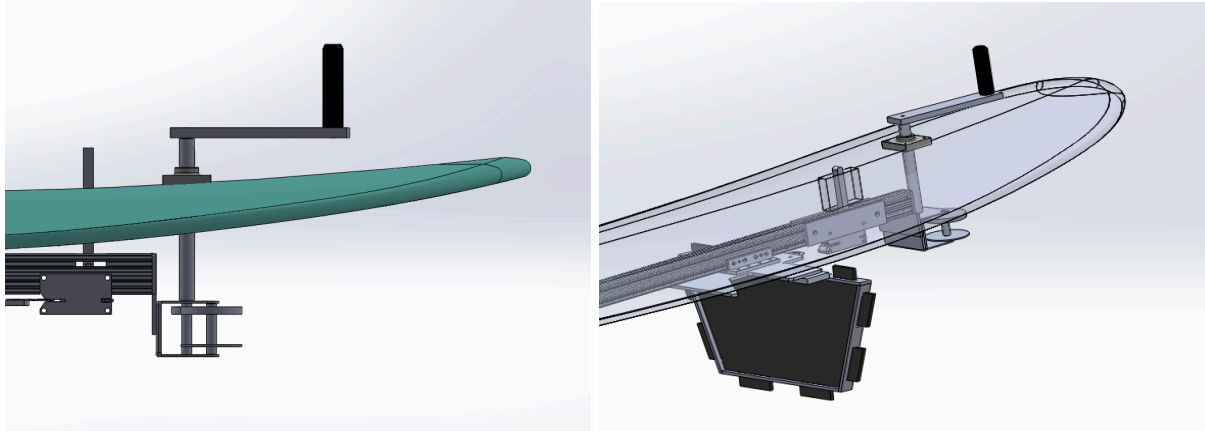


Figure 5: Full assembly with integrated trigger and winch system

This conceptual design showcases the propulsion system's evolution, addressing previous prototypes' limitations while introducing innovative solutions to improve performance, safety, and user-friendliness. By combining crossbow-inspired trigger technology, a high gear ratio winch, and a thoughtfully designed paddle mechanism, the system delivers an accessible and effective tool for surfers of all levels. The design enhances wave-riding capabilities and sets a new standard for safety and usability in surfboard propulsion systems. New paddle designs and rocker arms were dropped due to safety issues. The rocker arms had some potential to improve efficiency, but lacked a robust locking and triggering mechanism. The magnets likely lacked the strength and corrosion resistance to survive the harsh marine environment. Finally, the modified paddle designs were dropped due to their aggressive geometry, which may cause significant bodily harm to inexperienced users who misuse the device.

3. DETAILED DESIGN AND ANALYSIS

The surfboard propulsion system is divided into two primary subsystems: loading and propulsion, each designed to boost performance and ease of use. The propulsion subsystem is crucial in generating the thrust needed to help surfers catch waves. This system incorporates a paddle that applies force to the water, driven by high-stiffness springs that enable the paddle to move at a high velocity. The paddle assembly comprises a 0.25-inch neoprene rubber sheet and an aluminum frame to support the sheet. A rubber paddle was preferred to improve its safety since the rubber would deform and absorb energy upon human contact. Another benefit of using the neoprene paddle would be the potential to improve the drag coefficient of the paddle. As the paddle deforms, it takes on a concave shape, increasing the drag coefficient and improving the overall thrust capabilities. The propulsion system also includes a trolley that connects the paddle rides along a T-slotted rail, guiding the paddle along a linear path, ensuring smooth and consistent motion beneath the surfboard. Additionally, a hinge mechanism allows the paddle to transition seamlessly between a parallel and perpendicular orientation to the board, minimizing drag while the system is not being used and maximizing thrust when the system is fired. This design attempts to maximize the system's efficiency and functionality.

The loading subsystem is designed to reset the propulsion mechanism efficiently and easily. This system has a winch and trigger mechanism with a safety and ratchet switch. The winch allows the user to pull the paddle back into place after firing without much user effort. The

system is composed of several key components, which include a crank handle for manual loading, a driving and driven gear pair to achieve the desired gear ratio, a ratchet mechanism to prevent accidental backspin, a high-strength wet-environment Dyneema rope to pull the paddle into position, and a drum to store the rope neatly. Additionally, a carabiner was applied to the Dyneema rope, and an eyelet was placed on the trolley to avoid unnecessary winch rotation and endanger the user. This would allow the user to quickly and without much effort engage the winch to the trolley during loading and disengage it once complete. The loading process follows: attach the carabiner to the trolley, engage the ratchet, reel in the paddle until locked, and remove the carabiner from the trolley. This process can be completed without much effort while on the board.

The trigger mechanism is vital for securely activating the propulsion system, releasing the paddle from its locked position only when all safety protocols are met. It comprises a buckle, crossbow-inspired lock, safety, and trigger lever. To activate, several safeguards must first be disengaged: the safety tab from the crossbow trigger must be released, the ratchet on the winch must be disengaged, and the carabiner must be removed from the paddle. After these steps, the user can push the lever on the top of the board, which allows the trigger mechanism to release the buckle and set the paddle in motion along the track. These layered safety measures significantly reduce the likelihood of accidental misfires, ensuring a secure and reliable system for users.

We have conducted several engineering analyses on different propulsion system components to ensure the system is functional and efficient. One of the most critical areas we analyzed was the total force, as this directly impacts the overall speed achieved by the board. Using the strongest springs available to our team, we calculated the maximum spring force based on *Hooke's Law*. *Hooke's Law* states $F = k\Delta x$, where k is the spring constant and Δx is the displacement. With a spring constant of 2.14 lb/in, an elongation of 29.53 inches (75 cm) along the track, and two parallel springs, the total spring force is 126.38 lbs. Using the first derivative of this equation, we can find the total energy storage of the system. The spring potential energy can be found using the equation $U = \frac{1}{2}k\Delta x^2$. The total energy stored in our system was 1866 inch-pounds or 210.83 joules. We expect the paddle to move for a total duration of about 1 second. Thus, the average power output of the system would be 210 W.

After the maximum possible force output was obtained, our next action was to ensure that the user could comfortably load the system with the winch. To do so, we assessed the minimum appropriate mechanical advantage necessary to keep the input force below 15 lbs. To find an appropriate gear ratio to drive the winch, we used the formula $\frac{F_{out}}{F_{in}}$, where F_{in} is the input force and F_{out} is the maximum spring force. The results of our calculations are plotted in **Figure C.2**. Our results concluded that the winch system would need to provide a minimum of an 8:1 mechanical advantage. We considered these findings when buying the appropriate winch. A winch was purchased, and an analysis was conducted to determine the mechanical advantage. After considering the diameter of the selected rope, the mechanical advantage is expected to be 26. Calculations are provided in the appendix as **Figure C.2a**.

Given the maximum possible force output, it was important to determine if the load-bearing components would be strong enough to bear this load without deformation and failure. Therefore, a SolidWorks mechanical simulation was run on the trigger, given the realistic

maximum load of 126.4 lbs. An image of the simulation outcomes was provided in **Figure C.4**. The simulation predicted a maximum stress of 89 MPa. This stress level would exceed the maximum limit for some grades of aluminum, such as multipurpose aluminum 6061. Therefore, a higher grade of aluminum was chosen: aluminum 7075. This aluminum has a yield strength of 420 MPa, which is high enough to provide a 4.7 safety factor.

Another factor to consider was the efficiency's effect on the final velocity. Upon launch, the paddle must become perpendicular to the board to achieve high efficiency. However, there is a time when the paddle transitions from flush with the board to perpendicular. Because of this, some of the thrust is lost on startup. In addition, the cross-sectional profile of the board increased due to added components. This will result in increased drag and losses. Knowing that the system weighs 250 lbs with a user, we calculated the maximum theoretical velocity change through the equation $KE = \frac{1}{2}mv^2$, where m is the system's mass, v is the velocity, and KE is the kinetic energy. Since the energy stored in springs must be transferred to the environment, an assumption was made that the spring energy is equivalent to the kinetic energy plus losses. If we consider the kinetic energy at a maximum of 210.83 joules, the change in velocity compared to the mechanism's efficiency can be seen in **Figure C.3**.

The surfboard and surfer must travel between 3 and 4 meters per second to catch a small, slow wave. We expect most individuals to achieve a speed of 1 to 2 meters per second unassisted. However, the last 1 to 2 meters per second is a much more difficult task due to a non-linear increase in drag with an increase in velocity. **Figure C.3** illustrates that the system can function at efficiencies as low as 30% and still offer a 1-meter-per-second change in velocity. This is a promising outcome, which our group believes our prototype can accomplish.

A model of the paddle and the surfboard was also created in MATLAB using a non-linear ODE. First, a paddle model was made, and treated like a spring-mass-damper system, with the paddle being the mass and the drag force applied by the water being the damping element. However, the model was non-linear because the drag force was a function of the squared paddle velocity. The paddle system was modeled using the following equation: $mx'' + \frac{C_d}{2}\rho A(x')^2 + kx = 0$ where m refers to the mass of the paddle, ρ refers to the density of water, C_d refers to the drag coefficient of the paddle, and k is the combined spring constant of the propulsion springs. To simplify the model, the coefficient of drag of a flat plate, 1.28, was used. The model was then solved numerically and plotted against time in **Figure C.5**. Similarly, a model of the dynamics of the entire surfboard was made. Using data collected from drag testing, the drag coefficient of the surfboard was estimated to be 0.07389. The equation used to model the surfboard is $MX'' + \frac{C_{d,board}}{2}\rho A_{board}(X')^2 - F_{paddle}(t) = 0$ where M refers to the system's mass, and X refers to the displacement of the board. The final term $F_{paddle}(t)$ is the drag force that the paddle imparts, the second term of the paddle dynamics equation. Once again, this model was solved numerically and plotted against time in **Figure C.6**.

Finally, the neoprene paddle was simulated to observe the deformation and the stress applied to the rubber. Since neoprene exhibits hyperelastic properties, a two-term Mooney-Rivlin hyperelastic model was used. The simulation was conducted for several paddle speeds, the highest being 3 m/s. The stagnation pressure for this velocity in water is 4500 Pa, which was

calculated using Bernoulli's equation $\frac{1}{2}\rho v^2 = P_{stag}$. The resulting simulation informed the decision to purchase a neoprene rubber of 0.25 inch to reduce the stress on the paddle to a maximum of 1.6 MPa. This provides a safety margin of 6.25, considering that the tensile strength of the rubber was 10 MPa. An image of stress simulation is provided in **Figure C.7**

4. MANUFACTURING PROCESSES

A focus on functionality, cost-effectiveness, and sustainability drove the manufacturability of the surfboard propulsion system. Every part was selected or fabricated to meet the design specifications while considering the system's usability, durability, and \$700 budget constraints. By strategically reusing components from last year's prototype and carefully selecting additional materials, the team aimed to optimize performance without exceeding the budget or sacrificing quality. The decision to dismantle the previous prototype allowed the group to salvage essential components, reducing costs and streamlining the assembly process.

Key components of the system include the winch, high-strength springs, and crossbow-style trigger mechanism, along with secondary parts such as the rail-trolley system, paddle, carabiner, and rope. The winch plays a crucial role in simplifying reloading and increasing the total amount of energy stored. It is constructed primarily from stainless steel, a material chosen for its strength, long-lasting properties, and corrosion resistance in wet environments. The high-strength springs provide the force necessary to propel the paddle along the rail, while the crossbow-style trigger mechanism ensures safe and reliable operation under high-force conditions. The trigger is made from aluminum, a material selected for its durability, manufacturability, and low weight. Secondary parts, such as the rail-trolley system, facilitate smooth paddle movement, while the neoprene-aluminum paddle and Dyneema rope were chosen for their lightweight, durable, and corrosion-resistant properties.

The selection of new parts was carefully considered to enhance the system's functionality. Dyneema rope was selected for its strength and flexibility to serve as the cable for the winch. It is also designed for use in wet environments with a polyurethane coating for durability in high-friction applications. An aluminum dowel was made into a shaft to extend the winch through the surfboard, allowing the user to operate the reloading system from the top of the board. Additionally, essential hardware, including nuts, bolts, washers, and button heads, was used for the assembly. Finally, a highly durable carabiner was chosen to ensure that the entire load could be handled safely. These components are universally useful and essential for fastening parts together, particularly in a design where the environment is chaotic and unforgiving.

The group also identified opportunities to optimize reused parts to improve the design. For instance, the rail from the previous prototype along which the paddle traveled along, was modified to account for several parts of the new design. The mill was used to cut through the rail at certain areas, allowing for the trigger to have a slot within the rail, and for a lever to pass from the trigger to the top of the board. Additionally, the trolley system that carried the paddle along the track was able to be reused and modified to accommodate the new paddle. Weight considerations were central to the project, as adding too much weight could negatively impact

the surfboard's performance in the water. Therefore, aluminum was preferred over steel for most components, as it offers excellent strength-to-weight properties and superior corrosion resistance.

The original HDPE paddle was replaced with a new design featuring a neoprene paddle reinforced by an aluminum frame, selected for its superior flexibility and enhanced thrust potential. The frame was fabricated from one bar of aluminum that was cut down to four pieces, given groove cutouts, and then welded together at the vertices, giving it a trapezoid shape. The neoprene was cut into a similar trapezoidal shape with tabs along the edges, which were inserted into the groove cutouts to anchor the material securely using tension and pressure. This redesigned paddle was modified with the existing trolley system to be able to translate smoothly from one end of the rail to the other. To boost the energy output of our system, the original tension springs were replaced with high-power tension springs boasting double the resistance of the originals. Through the mechanical advantage of the system's winch, these high-power tension springs would provide an achievable performance boost without any added difficulty when it came to loading.

One of the major subsystems, upon which the safety and success of the main system hinged, was the trigger. Our custom trigger was inspired and modeled after a simple crossbow trigger consisting of 3 parts, depicted in **Figure C.9**. From left to right in **Figure C.9**, one can see a latch to lock our mechanism, a handle to release the latch, and a safety to prevent the handle from moving when the device should not be engaged. Both the latch and the handle are positioned in such a way as to support small springs to make the trigger subsystem spring-loaded and capable of returning to a default state for loading. The safety within the trigger subsystem is specifically not spring-loaded and must be moved manually to prevent and allow the handle to function. Unfortunately, moving the safety in its engaged or disengaged position does not allow it to stay in one position. A minor solution to manually moving the safety came in the form of an elastic band, which, when positioned on the left side of the safety, forces it to be disengaged and, when positioned on the right side of the safety, forces it to be engaged. This proved to be useful and is something that can be improved upon in future iterations of this subsystem.

After performing simulations of maximum stress and tension with this trigger subsystem, it became apparent that the weakest part within our design was the latch that allows the paddle to lock into the mechanism. The simulation for that latch specifically is shown in **Figure C.4** using a Von Mises style analysis. According to this analysis, the maximum amount of stress this single piece could handle before deformation was 89 megapascals. With this maximum stress in mind, the team determined that for the latch part of the trigger subsystem, a stronger grade of aluminum would be required to ensure that the part did not break under optimal and maximum conditions. For this, 7075 grade high-strength aluminum was purchased and used for the latch part as it provided the necessary capabilities to handle the simulated maximum stress.

The manufacturing of the trigger subsystem itself proved to be a challenging task, considering the pieces that made up the trigger needed to be small and very precise. Because of this, the Surfboard Propulsion team contacted the head of the machine shop on campus and worked with him closely to create the subsystem. The housing around the individual pieces was created first using an aluminum block and a mill to cut away material at half depth. This ensured that the bottom half of the aluminum maintained its solidity and yet still allowed for a concavity

in which to house the individual pieces. After the housing was created, the following smaller parts were made with a combination of milling techniques and CNC machining techniques to create the desired shape parameters. Once the housing, latch, handle, and safety parts were all individually made, they could be laid out within the subsystem and marked for pin hole placement. The holes would create points of interest where pressure-inserted pins could be used to allow the parts to move between themselves without the risk of falling out of the housing. To finish the machining of the trigger mechanism, an acrylic plate was made to enclose the subsystem and allow users of the device to see the individual parts moving in real time.

Assembly of the trigger mechanism, while thought to be easy, proved to pose small problems, specifically the clearances between parts. These challenges were easily rectified by shaving down small fractions from some parts while maintaining strength and overall shape. Once the parts were fitted into the housing and posed no issues, the pressure-fitted pins were inserted through each part and the housing. Afterwards, the springs were snugly placed within specific grooves in the housing, and the acrylic front plate was secured in place. This system, while sleek and elegant, poses some potential issues after minor use, specifically when considering the placement of the springs in the housing. The springs were usable and consistently returned the individual parts to their original positions, but would occasionally fall over or become misaligned slightly. Therefore, the grooves of the springs must be altered or changed in some way in future considerations of this mechanism to ensure spring stability.

Issues arose when testing the trigger mechanism and its ability to lock to the buckle. At times, the buckle would be misaligned from the trigger hook, allowing the trolley to move freely. This would cause undesired launches if the ratchet were removed. To remedy this, the inside of the buckle was filed down to create a larger area for the hook to latch on to. This way, even if the buckle were to come in at a slight angle, the trigger would be able to latch on properly and safely hold the trolley in place.

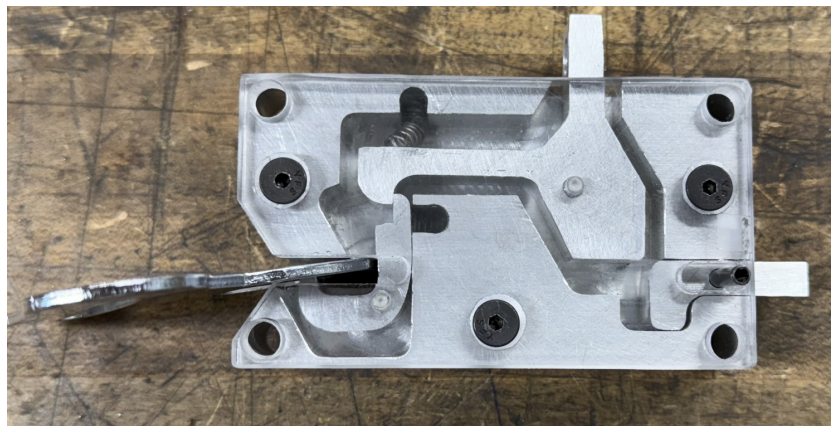


Figure 3: Trigger Mechanism

Other smaller manufactured parts of the system included the shaft-crank, lever handle with grip, and mounting plates for the winch. The shaft was made from aluminum and designed to connect to the driver axle of the winch. Initially, it was threaded to match the threads on the steel driver axle, allowing it to be twisted into place. However, this approach led to mechanical

failures, as the aluminum threads on the shaft would strip due to the hardness difference between the two metals. To resolve this issue, the threaded connection was replaced with a more robust solution: a cut was made into the shaft, and a set screw was used to securely fasten it to the axle.

Manufacturability was a critical consideration in the design process. Parts were selected to be easily fabricated using standard manufacturing techniques such as CNC machining and traditional milling. The trigger mechanism, for example, was machined with precision using CNC milling to meet the required tolerances for safe operation. Aluminum extrusions were chosen for the rail system due to their ease of customization and durability. This approach minimized production costs and simplified assembly, making the design feasible for both prototyping and large-scale production. Additionally, sustainability played an essential role, with corrosion-resistant and recyclable materials prioritized to reduce maintenance needs and minimize environmental impact over the system's lifespan.

The choice of parts and materials influenced several aspects of the final design. The winch system, for instance, was preferred over a block-and-tackle mechanism due to its simplicity and ease of modification. This decision required adjustments to the housing and mounting components to ensure seamless integration with the rail-trolley system. Similarly, transitioning to a crossbow-style trigger improved safety and usability, necessitating the locking mechanism and associated housing refinements. These changes ultimately enhanced the system's performance while maintaining ease of production.

Several challenges were faced during manufacturing considerations, including restrictions on purchasing parts from unapproved vendors. While the team identified alternative suppliers offering lower prices, approval from the TA was required to ensure compliance with school regulations. Salvaging parts from the old prototype proved to be an effective solution for reducing costs and lead times, as these components did not require shipping or additional procurement steps. Another challenge involved coordinating the fabrication of critical components, such as the trigger mechanism. Recognizing the importance of this part, the team prioritized finalizing the design early and sought assistance from Mr. Ioan-Mihai Gradina to ensure its precise and timely manufacturing. Submitting the trigger design early was essential to ensure timely delivery, as the fabrication queue typically becomes crowded later in the semester.

In summary, the manufacturability of the surfboard propulsion system reflects a thoughtful balance of functionality, sustainability, and cost-effectiveness. By reusing parts where possible, selecting lightweight and durable materials, and leveraging standard manufacturing processes, the team was able to create a design that is scalable and practical. The project is committed to performance and usability while adhering to budget and regulatory constraints. For a detailed breakdown of the parts and costs, refer to Appendix B.

5. TESTING

Other than having a reliable and safe structure, the goal of the device was to displace water and accelerate the paddle and, consequently, the board. Previously, dynamic models were created to predict the motion of both the paddle and the board based on parameters such as mass, spring constant, and drag coefficients. The goal of testing was to collect position data of the

paddle and the board in two separate tests. Then this data is plotted alongside the dynamic models. Thus, the motion of the paddle and board can be compared with the model. This can allow the model's constants, such as the drag coefficient, to be modified to fit the data. Additionally, velocity was extracted from the position data using numerical differentiation.

The data collection process included using a GoPro Hero slow-motion camera and MATLAB's video and image viewer software. The paddle was fired with the board placed in water and held to prevent any motion. The GoPro camera recorded the paddle's motion relative to the board at 120 fps. The captured footage was analyzed by measuring the pixel distance of the paddle relative to its ending position and then converting the pixel distance into standard distance units. This distance was measured for every seven frames and plotted concerning time since the frame rate was known. Similarly, the GoPro camera recorded the board's motion relative to the ground, yet the distance was measured every nine frames due to the slightly extended duration of the test.

Below are **Figures C.8** and **C.9**, which are plots of the paddle and board's dynamics, respectively. The collected data is shown side by side with the model. **Figure C.8's** model uses a drag coefficient of 1.28 (flat plate) for the paddle. However, the collected displacement data suggests that the drag coefficient of the paddle must be greater since the real response of the paddle is more damped. This confirms the idea that a curved, rubber paddle would be more effective at producing drag. This is because drag is the damping component of this spring-mass-damper model. A drag coefficient of 1.75 fits the displacement data far better, suggesting that a rubber paddle that can deform is an added benefit. **Figure C.9's** model is modified compared to **Figure C.6**. The original model in **Figure C.6** uses a surfboard drag coefficient of 0.09, which was fitted for the previous prototype's drag coefficient. However, adding a winch and trigger mechanism has increased the overall drag coefficient. Therefore, **Figure C.9** was modified to include a drag coefficient of 0.12 rather than 0.09, and it utilizes a paddle drag coefficient of 1.75 as previously stated. However, these changes still don't reflect a great fit with the data.

The discrepancy between the board data in **Figure C.9** is likely due to the nose of the board jumping up out of the water for a brief period right after the trigger is activated. The high velocity of the board during the first 0.3 seconds is due to the rapid weight transfer from the front to the rear of the board when the nose of the board is in the air and drag is reduced. This error could be explained by the lack of a user on the board during testing. This was done to simplify the image capture process and be consistent with the model's previous parameters.

All in all, the design met our model's expectations within a reasonable degree of error. Although testing was done without a user, it is clear that the design can generate thrust safely and in a controlled manner. It is important to note that power is only delivered for a very short time (less than one second), making the device challenging to use. Future prototype iterations can focus more on balanced power delivery since many safety measures have been applied to the board.

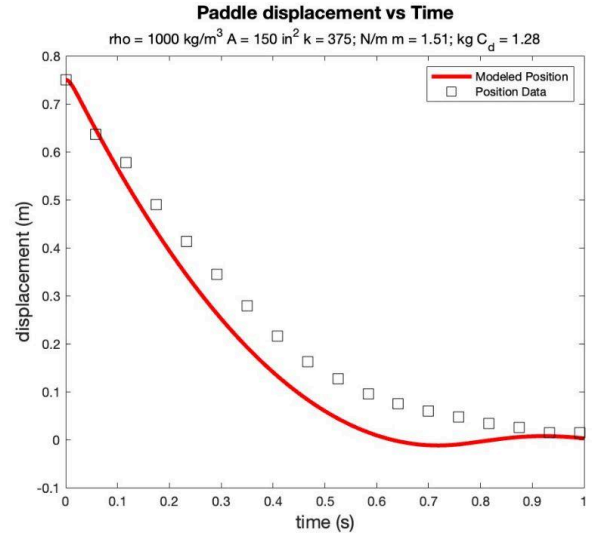
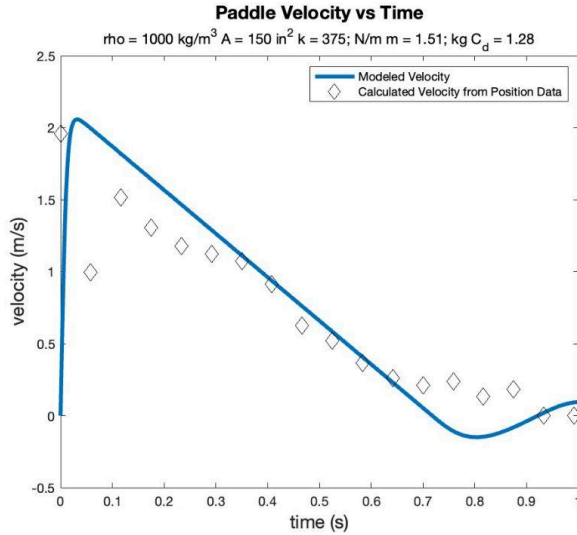


Figure C.8: Paddle Model and Data Dynamics

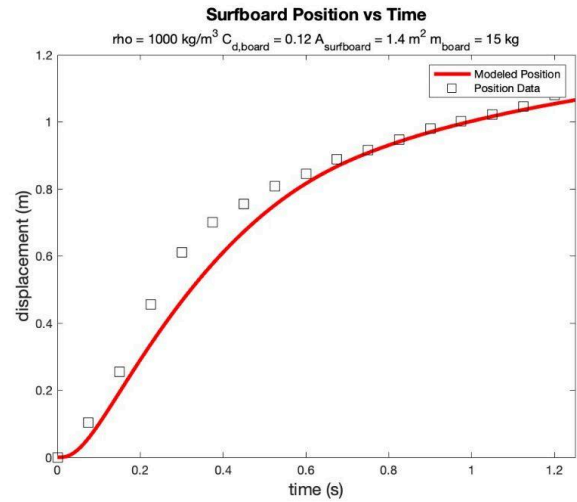
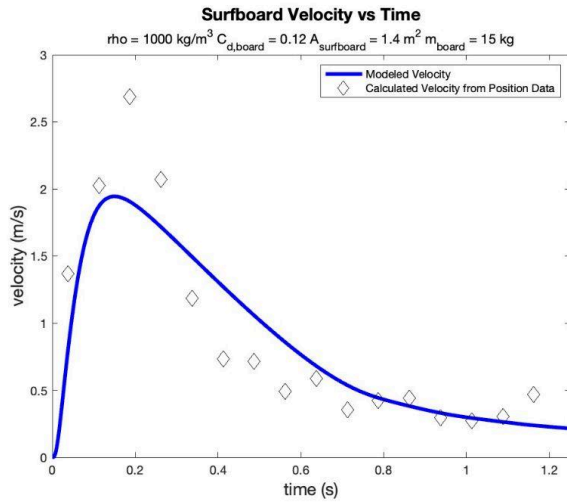


Figure C.9: Board Model and Data Dynamics

6. CONCLUSION

The team improved efficiency, safety, and user convenience by designing an enhanced surfboard propulsion system. The system features a winch, a custom-designed crossbow-inspired trigger, higher-strength springs, and integrated safety mechanisms, providing thrust through a neoprene-aluminum paddle that effectively displaces water. Iterative design improvements addressed critical challenges such as energy storage, safe deployment, and user-friendliness, resulting in a robust and reliable solution.

Initial testing showed the propulsion system's ability to provide consistent thrust and safe operation under controlled conditions. It also gave insight into the maximum speeds the board could reach, which was around 2.5 m/s from a standstill. This proved to be higher than the

previous prototype. Moving forward, more testing will be necessary to see if the board can handle saltwater environments.

While the current design meets most project objectives, future iterations could include improvements such as optimizing the paddle for better water displacement, adding advanced safety features to optimize our safety mechanism, adding a simpler method to disengage the ratchet, and increasing the gear ratio of the winch to allow for even higher energy storage. These adjustments will enhance the system's efficiency, reliability, and safety.

This project highlights the potential for innovative mechanical solutions to make surfing more accessible and enjoyable. By balancing functionality, safety, and sustainability, the surfboard propulsion system lays a solid foundation for advancing surfing technology and expanding the sport's inclusivity to a broader audience.

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Appendix A. PROJECT MANAGEMENT

This section outlines the project management approach our team used throughout the development of the surfboard propulsion system, from initial research to construction and testing. The project was structured across two semesters: Fall 2024 focused on concept development and design planning, while Spring 2025 emphasized manufacturing, testing, and refinement. In the Fall, we began by identifying the limitations of existing propulsion systems and brainstorming improvements. This phase involved defining the problem, evaluating potential concepts, and iterating on designs through CAD modeling and group discussions. Major deliverables included the Critical Design Report, Final Presentation, and, most importantly, establishing a clear and finalized vision of what our propulsion system would look like and how it would function. Another key requirement was producing a complete and accurate parts list before the end of the semester to ensure a smooth transition into the spring manufacturing phase. These milestones allowed us to close the semester with a solid design foundation and a clear direction for execution.

One of the most significant challenges during the Fall semester was managing evolving expectations. Often, when we believed we had settled on a design or were ready to begin ordering components, our advisors would raise concerns—either that the design was overly complex or that it didn't align with their vision. This led to repeated revisions, delays in progress, and the need to reevaluate ideas we had already invested time in. While this was frustrating, it taught us the importance of flexibility and persistence. We learned to manage our timeline with greater care, balancing advisor feedback with our technical goals, and progressing in areas where we had clarity while continuing to refine uncertain elements.

As we moved into Spring 2025, our focus shifted to hands-on work: ordering parts, machining components, verifying tolerances, assembling the system, and testing its functionality. This phase is still in progress, but several foreseeable challenges have already emerged. One concern is the combined weight of the propulsion mechanism and the rider, which may affect the board's performance. We've also experienced misalignment issues with the seatbelt mechanism—something we're currently working to resolve to improve comfort and safety. Additionally, the trigger mechanism, while functional, may require further refinement to enhance its safety and reliability in real-world surf conditions.

To address these concerns, we're exploring the use of lighter materials to reduce overall weight and improve board handling. Enhancing the trigger mechanism for safety and ease of use is a priority, as is improving the paddle design to deliver more effective thrust. These future resolutions are part of our continued effort to optimize the system and ensure it performs consistently and safely under repeated use. We view this phase not only as validation of our current design, but also as an opportunity to evolve it further based on testing outcomes.

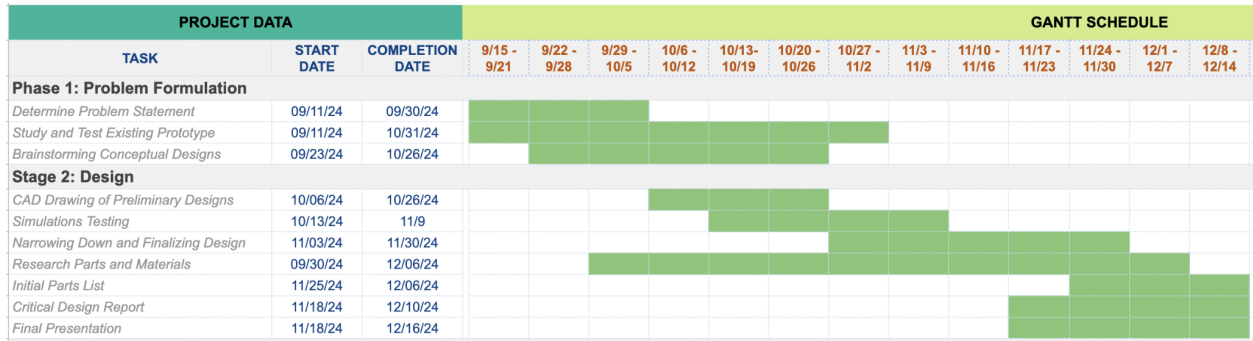


Figure A.1: This chart outlines the design and planning phase of the project, including concept development, CAD modeling, and finalizing the parts list. It was used to successfully track progress and ensure all major deliverables were completed on time before entering the manufacturing stage.

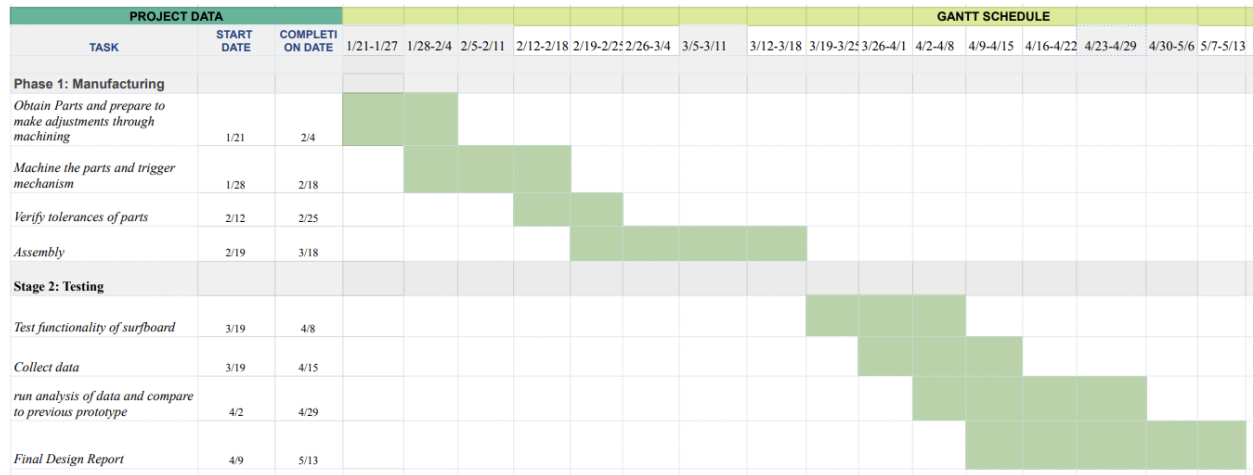


Figure A.2: This chart maps out the manufacturing, assembly, and testing phases of the project. It provided a structured timeline that guided our team through each stage, allowing us to complete fabrication, system integration, and validation within the planned schedule.

Appendix B. PARTS LIST & BUDGET

The Surfboard Propulsion Unit team had a working budget of \$700 provided by the Rutgers University Mechanical and Aerospace Engineering Department, with no external funding included. Our team went through four parts ordering waves and spent a total of \$325.14 to finish our project, leaving us with a surplus of \$374.86. McMaster was the sole vendor that was ordered, as it had all the parts we needed readily available.

Order Number	Vendor	Part Name/Description	Part ASIN/ Part Number	Quantity	Unit Price	Total Price
1	McMaster	T-slotted Framing Double Six Slot Rail	47065T107	1	\$73.91	\$73.91
1	McMaster	T-slotted Framing End-Feed Double Nut	47065T147	1	\$6.12	\$6.12
1	McMaster	T-slotted Framing End-Feed Single Nut	47065T139	1	\$3.05	\$3.05
1	McMaster	Wet-Environment Easy-Splice Rope	4377N1	1	\$0.63/ft	\$15.75
2	McMaster	High-Strength 7075 Aluminum Bar	9055K43	1	\$35.61	\$35.61
3	McMaster	Flange-Mounted Shaft Support	62645K42	1	\$24.19	\$24.19
3	McMaster	Aluminum Carabiner	31375T18	1	\$9.55	\$9.55
4	McMaster	High-Strength Multipurpose Neoprene Sheet	1290N57	1	\$83.07	\$83.07
4	McMaster	Multipurpose 6061 Aluminum Bar (2 ft)	8975K596	1	\$7.79	\$7.79
4	McMaster	Multipurpose 6061 Aluminum Bar (1 ft)	8975K596	3	\$4.18	\$12.54
4	McMaster	Aluminum Unthreaded Through-Hole-Mount Crank	6473K91	1	\$26.00	\$26.00
4	McMaster	302 Stainless Steel Corrosion-Resistant Compression Springs (0.75")	1986K303	1	\$6.88	\$6.88
4	McMaster	302 Stainless Steel Corrosion-Resistant Compression Springs (0.375")	1986K106	1	\$6.88	\$6.88
4	McMaster	18-8 Stainless Steel Dowel Pin	90145A474	1	\$13.80	\$13.80

Table B.1: Parts list and spending activities of your project.

Appendix C. DRAWINGS/ SIMULATION/CODES/DATA/DIAGRAMS

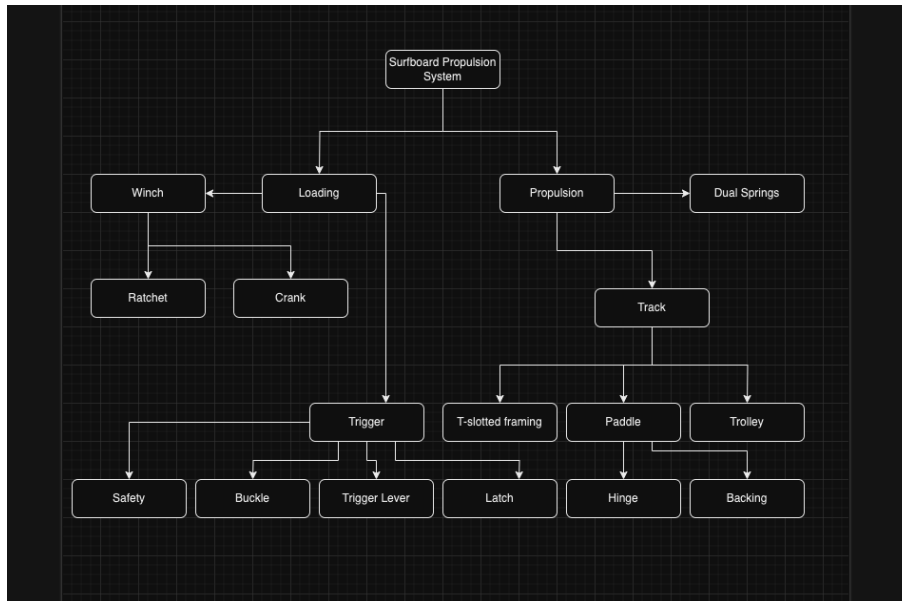


Figure C.1: Functional Decomposition

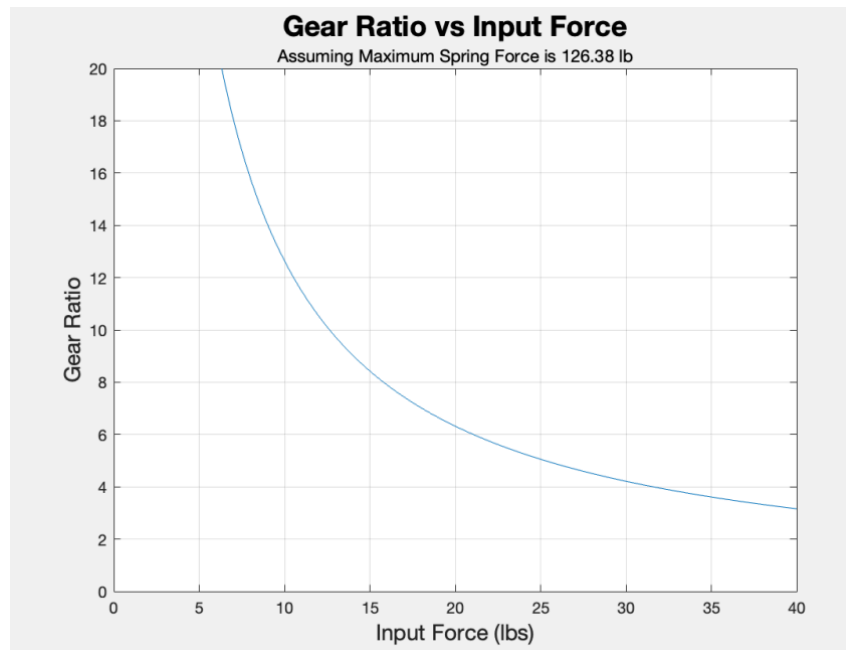


Figure C.2: Gear Ratio vs. Input Force

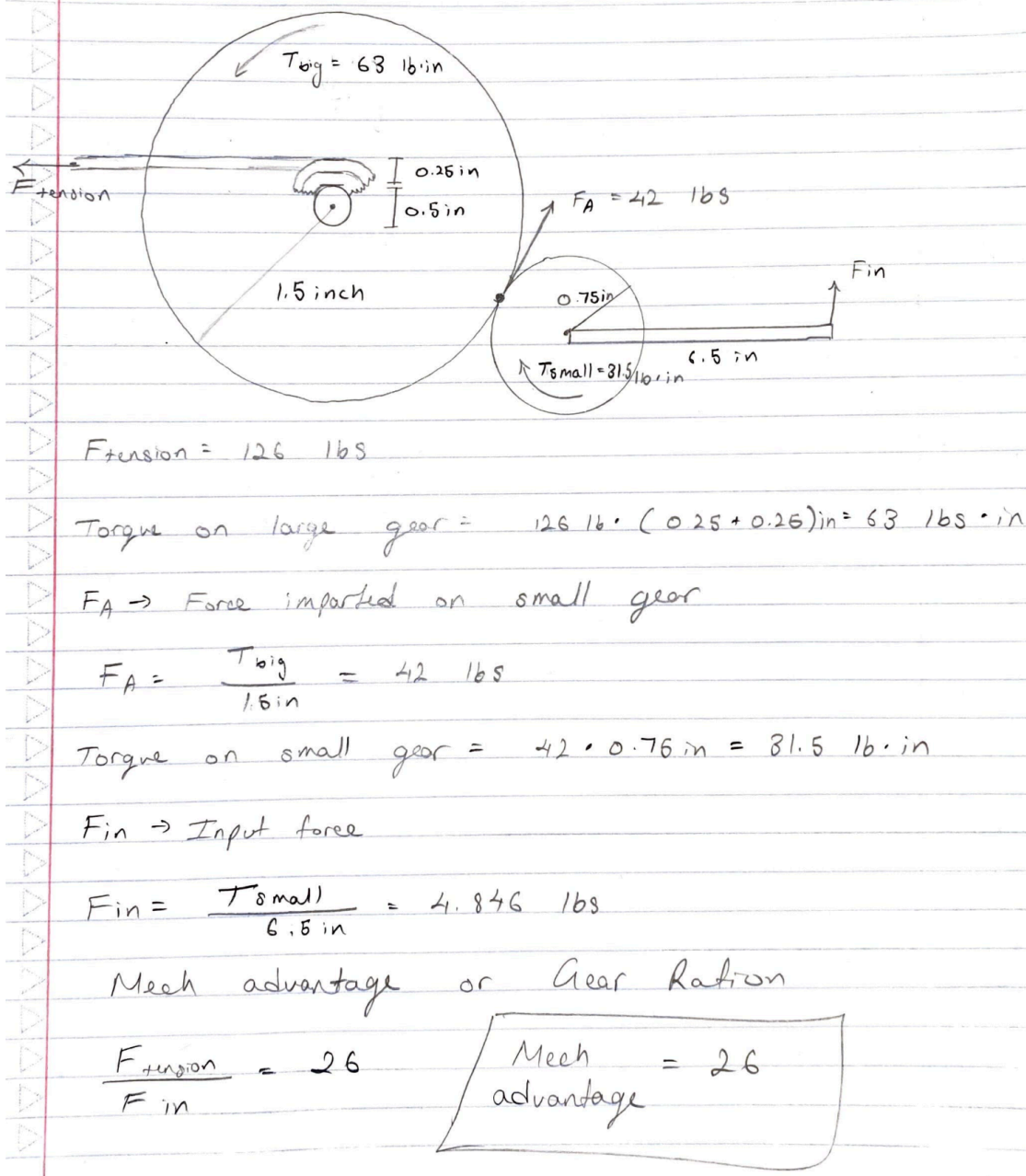


Figure C.2a: Mechanical Advantage Calculation

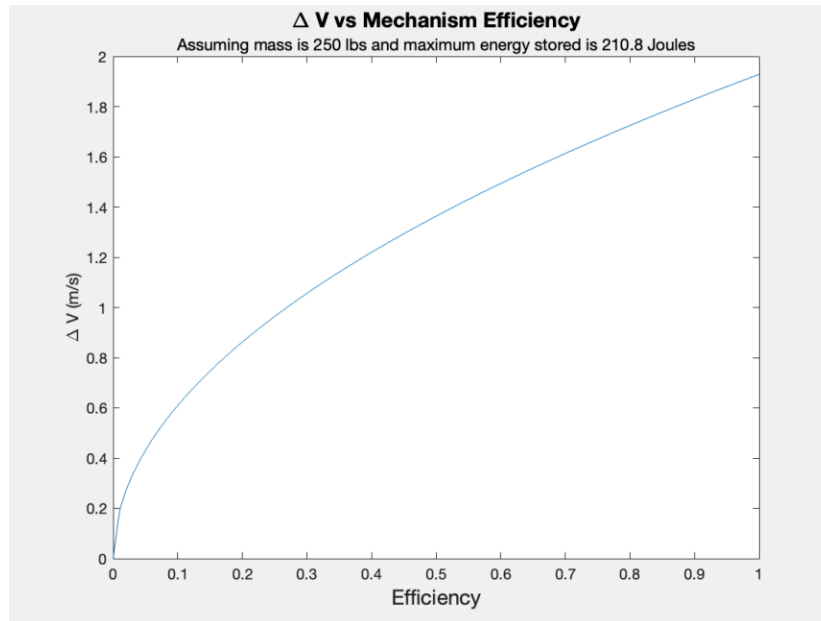


Figure C.3: Velocity vs. Mechanism Efficiency

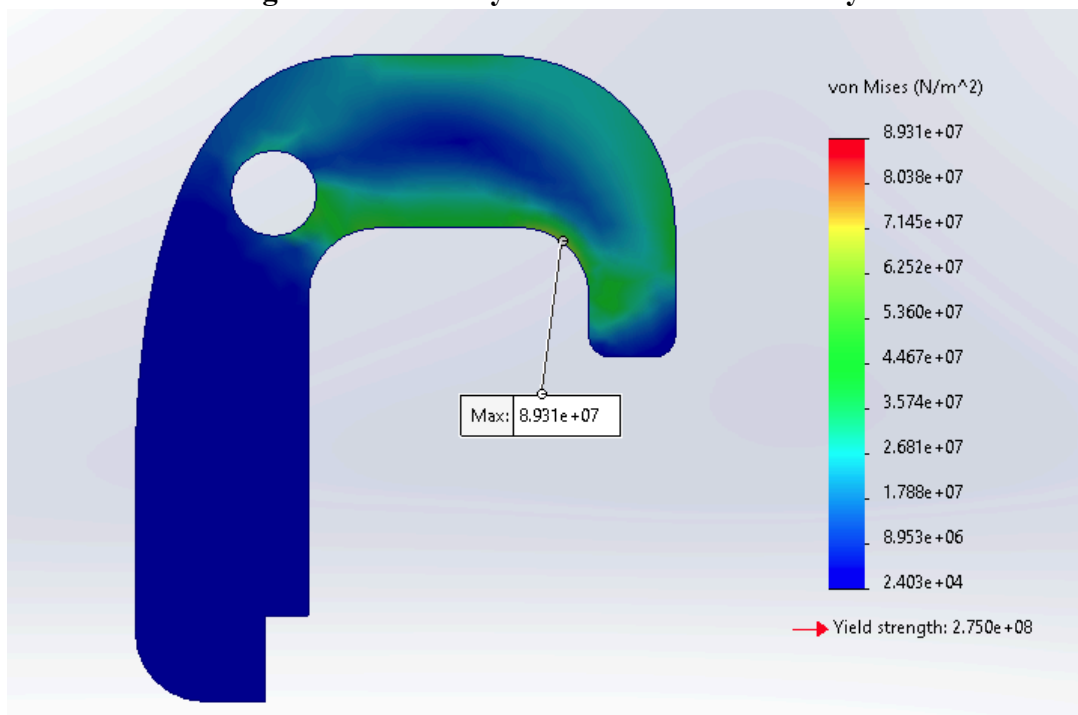


Figure C.4: Von Mises Stress of Load Bearing Trigger Component

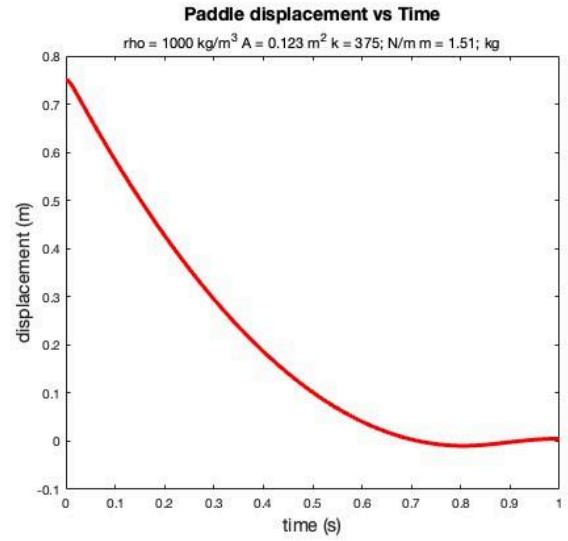
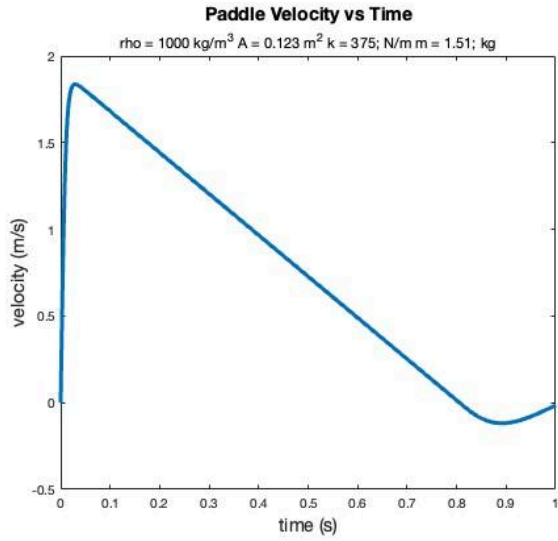


Figure C.5: Modeled Paddle Velocity and Displacement with zero initial velocity

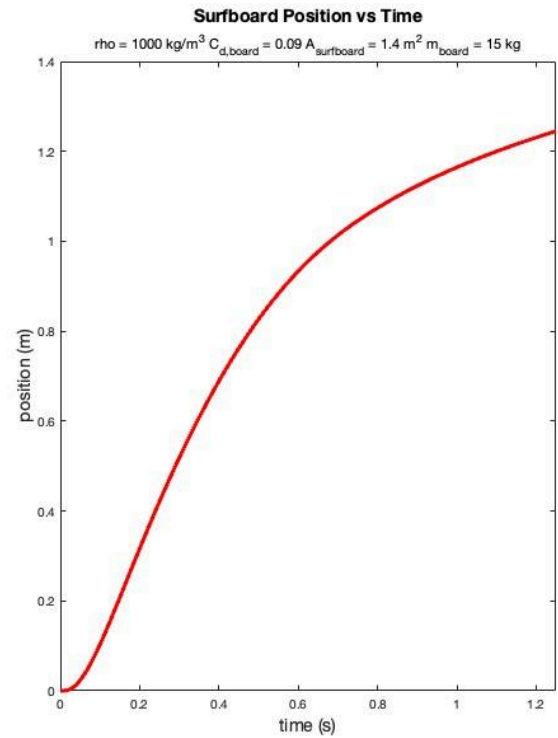
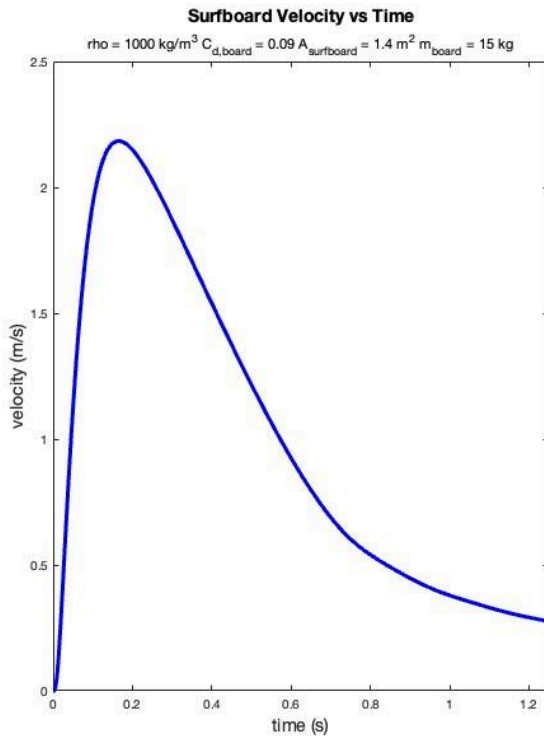


Figure C.6: Modeled Surfboard Velocity and Displacement with zero initial velocity

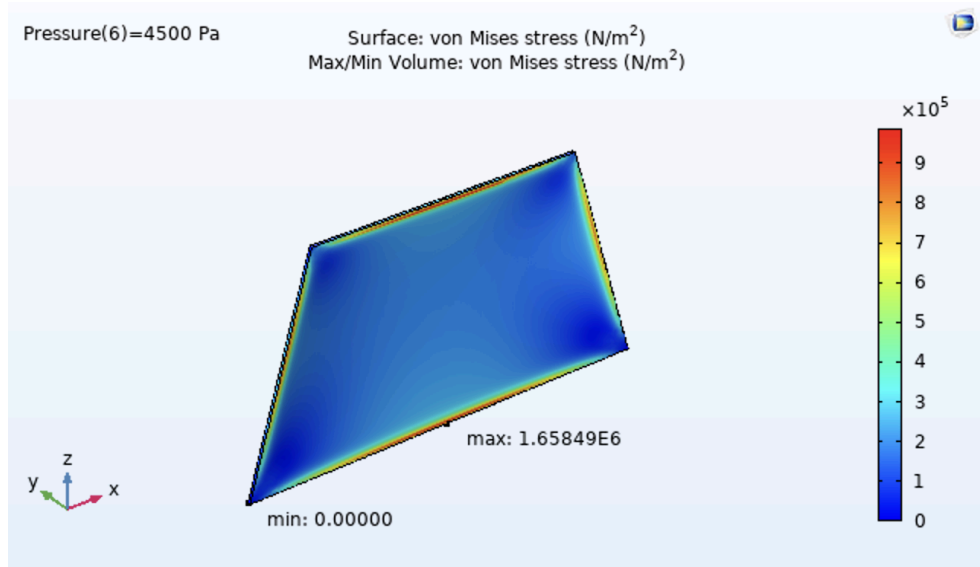


Figure C.7: Comsol Sim of Neoprene Rubber Sheet

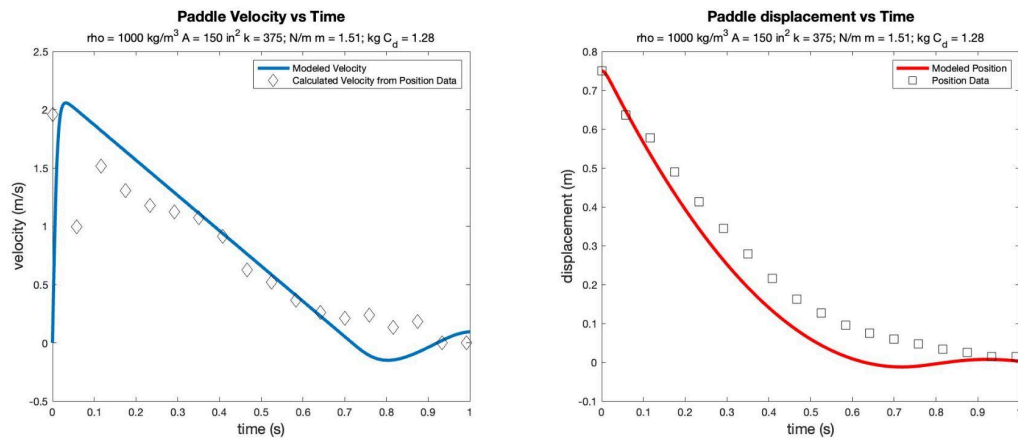


Figure C.8: Paddle Model and Data Dynamics

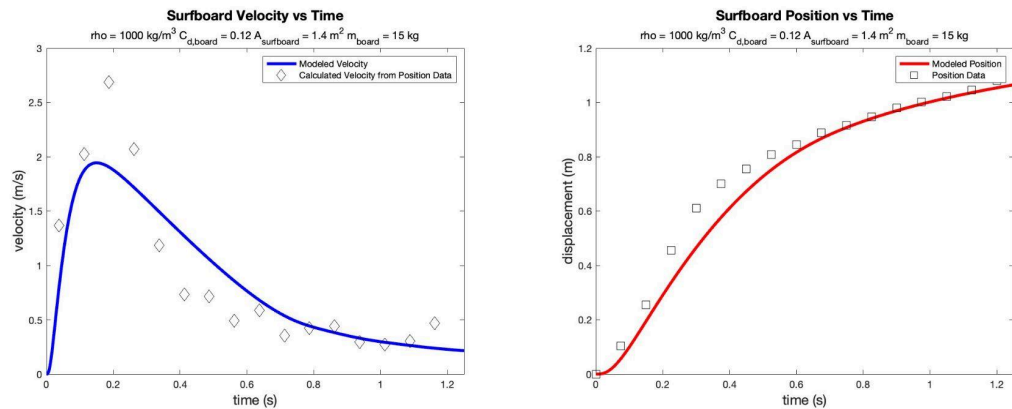


Figure C.9: Board Model and Data Dynamics

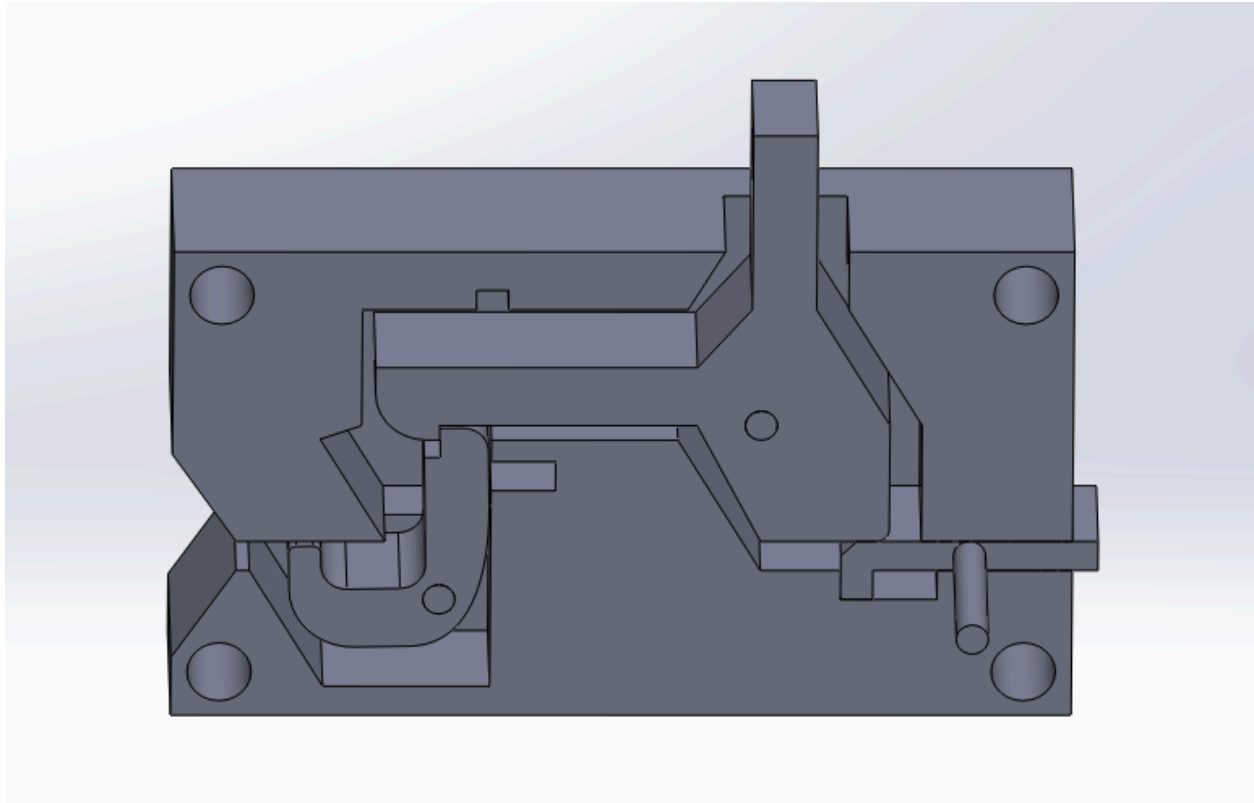


Figure C.10: Trigger Mechanism

```
function xdot = smd(t,x,rho,m,A,k)
% Nonlinear spring-mass-drag system function
xdot(1) = x(2);
%xdot(2) = -81.46*x(2)*abs(x(2))-248*x(1);
xdot(2) = (1.28*0.5*(-rho)/m) * A * x(2) * abs(x(2)) - k/m * x(1);
xdot=xdot';
```

Figure C.11: Spring Mass Damper Function

```

%% Defining Conditions of Spring-Mass-Damper System

rho = 1000; %kg/m^3
A = 0.096774; %m^2
k = 375; %N/m
m = 1.51; %kg
v0 = 0:0.25:2;

%% Model
[t,x] = ode45(@(t,x) smd(t,x,rho,m,A,k),[0 1],[0.75 v0(1)]);

%% Real Data

fps = 120; %frame/sec

startframe = 203; %frame number
endframe = 336; %frame number

start2end = 336-203; % total number of frames between start to finish

duration = start2end/fps;

zero_dis = 66; %starting distance between trigger and trolley

pixeldis = [66 141 179 237 287 332 375 416 451 475 495 509 519 527 536 541 548 548 548 548]; %distance between trigger and trolley
pixeldisincorrect = pixeldis-66; %standardize pixel distance
framecount = [203 210 217 224 231 238 245 252 259 266 273 280 287 294 301 308 315 322 329 336]; %which frame the pixeldis was taken
framecount = framecount-203;

realx = 0.75 - (0.86/564*pixeldisincorrect);
realt = framecount/fps;

realv = diff(realx)./diff(realt);

derivativeT = zeros(1,19);
for i = 1:19
    derivativeT(i) = realt(i);
end

```

Figure C.12: Paddle Simulation Script

```

function xy_dot = Surfboard_Motion_ODE(t, xy, rho, A_paddle, Cd, A_surfboard, m_paddle, m_board, k)

x = xy(1); %Paddle pos
xdot = xy(2); %Paddle velocity
y = xy(3); %Board pos
ydot = xy(4); %Board velocity
F = (1.75* 1000 * 0.5 * 0.123 * xdot.^2);
% Paddle ODE: ((p*A*(x')^2)/m) - x*k/m
xddot = (0.5*1.28 * (-rho)/m_paddle) * A_paddle * xdot * abs(xdot) - k/m_paddle * x; %Paddle Acceleration

% Surfboard ODE:
yddot = (F / m_board) - ((0.5 * rho * A_surfboard * Cd * ydot * abs(ydot))/ m_board); % Acceleration of Board

% Return the derivatives as a single column vector
xy_dot = [xdot; xddot; ydot; yddot];
end

```

Figure C.13: Combined Paddle/Board Function


```

% Params
rho = 1000; % Density of water (kg/m^3)
A_paddle = 0.096774; % Paddle cross-sectional area (m^2)
A_surfboard = 1.4; % Surfboard cross-sectional area (m^2)
m_paddle = 1.5; % Paddle mass (kg)
m_board = 15; % Surfboard mass (kg)
k = 375; % Spring constant for the paddle (N/m)
Cd = 0.09; %Coefficient of Drag

%Initial Conditions
x0 = 0.75; % Initial paddle position in meters
xdot0 = 0; % Initial paddle velocity (adjust if needed)
y0 = 0;
ydot0 = 0;
initials = [x0; xdot0; y0; ydot0];

% ODE SOLVER
[t, xy] = ode45(@(t, xy) Surfboard_Motion_ODE(t, xy, rho, A_paddle, Cd, A_surfboard, m_paddle, m_board, k), [0,1.25], initials);

```

Figure C.14: Combined Paddle/Board Script